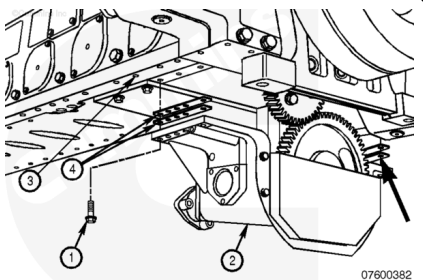


Install

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Install the pump (2) and shims (4) on the engine (3).

Install the mounting capscrews (1).

Tighten the capscrews.

Torque Value:

M12

1. 80 n•m [59 ft-lb]

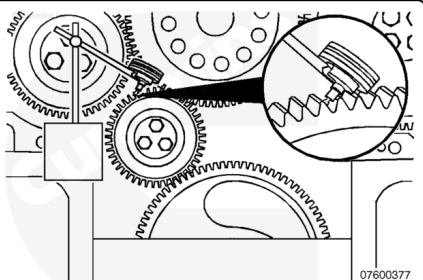
Torque Value:

M10

1. 45 n•m [33 ft-lb]

⚠ CAUTION ⚠

Backlash measurement is critical. If not properly adjusted, lubricating oil pump or engine failure will occur.



It is required to measure the lubricating oil pump backlash three times. It **must** be measured on the inverted engine when the pump is mounted, after the pump plumbing is installed, and again after the engine is rotated upright.

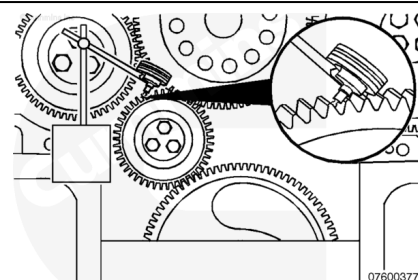
Lubricating Oil Pump Idler Gear Backlash		
mm		in
0.21	MIN	0.008
0.39	MAX	0.015

If the backlash is **not** within specification, adjust the lubricating oil pump.

Adjust

⚠ CAUTION ⚠

Backlash measurement is critical. If not properly adjusted, lubricating oil pump or engine failure will occur.



It is required to measure the lubricating oil pump backlash three times. It **must** be measured on the inverted engine when the pump is mounted, after the pump plumbing is installed, and again after the engine is rotated upright.

Use a dial indicator to measure the lubricating oil pump idler gear to crankshaft gear backlash.

Backlash - Lubricating Oil Pump Idler Gear to Crankshaft Gear

mm		in
0.21	MIN	0.008
0.39	MAX	0.015

If the backlash is less than specifications, remove the pump and install shims between the pump and the engine. Shims are available 0.13 mm [0.005 in], Part Number 4015782, and 0.25 mm [0.010 in], Part Number 4015783.

If the backlash is greater than specifications, remove any shims, and measure the backlash again. If no shims were used, the pump drive gear or the crankshaft gear is worn and **must** be replaced.

An equal amount of shim thickness **must** be used under both mounting feet, if shims are required.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Position the shims (4) between the pump (2) and the engine block (3).

Install the pump (2) and shims (4) on the engine with the six capscrews (1).

Tighten the capscrews.

Torque Value:

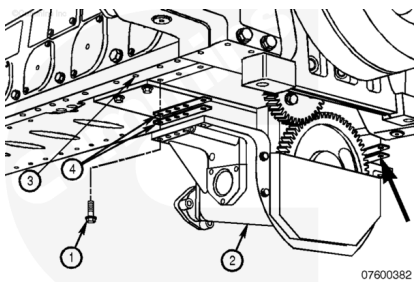
M12

1. 80 n•m [59 ft-lb]

Torque Value:

M10

1. 45 n•m [33 ft-lb]



Check the lubricating oil pump backlash.

Lubricating Oil Pump Idler Gear to Crankshaft Gear
Backlash

mm		in
0.21	MIN	0.008
0.39	MAX	0.015

